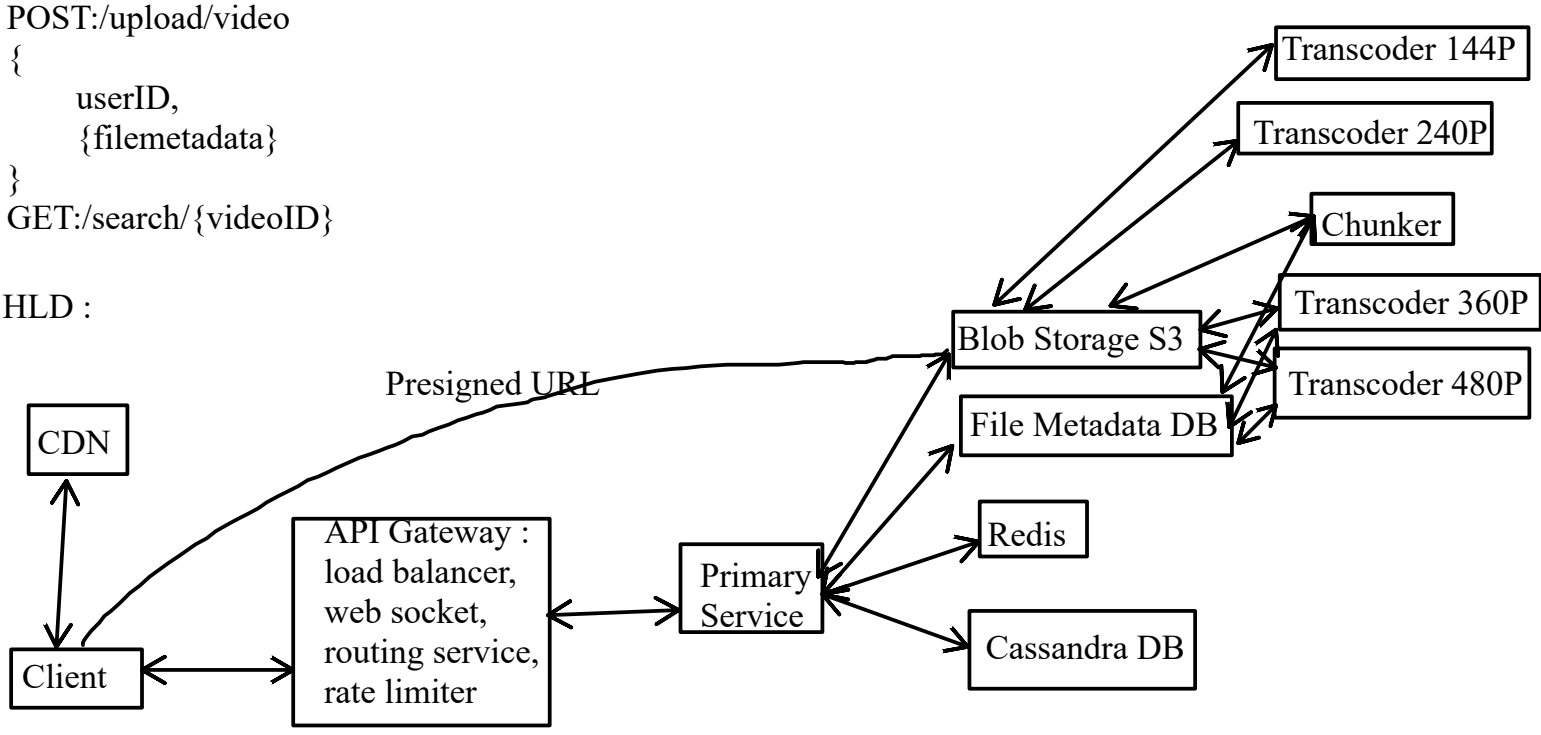
Youtube

Functional requirements:
User should be able to open/download any video based on his preference,
User should be able to download any videos based on his preference,
User should be able to upload any file >50GB

Non-functional requirements:
data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:
User ,
video,
history,
filemetadata

API :
POST:/upload/video
{
userID,
{filemetadata}
}
GET:/search/{videoID}

HLD :